Qatar-1b

R-0653

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-1_220615` · created 2026-07-25T23:46:20+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459745.8261 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.01
Transit depth (Rp/Rs)^2	3.7 +/- 0.4 [%]
Orbital Inclination (inc)	85.33 +/- 1.22
Airmass coefficient 1 (a1)	1.184 +/- 0.011
Airmass coefficient 2 (a2)	-0.114 +/- 0.036
Scatter in the residuals of the lightcurve fit is	0.95 %
Best Comparison Star	#1 - [131, 457]
Optimal Method	PSF photometry
Transit Duration (day)	0.0493 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.01 vs 0.1463 (deviation 3.72σ)
Duration fit vs published	0.0493 d vs 0.0692 d (deviation 4.89σ)
Mid-time O–C	3.6 min
Depth SNR	15.6
Residual scatter	0.95 %

Light curve

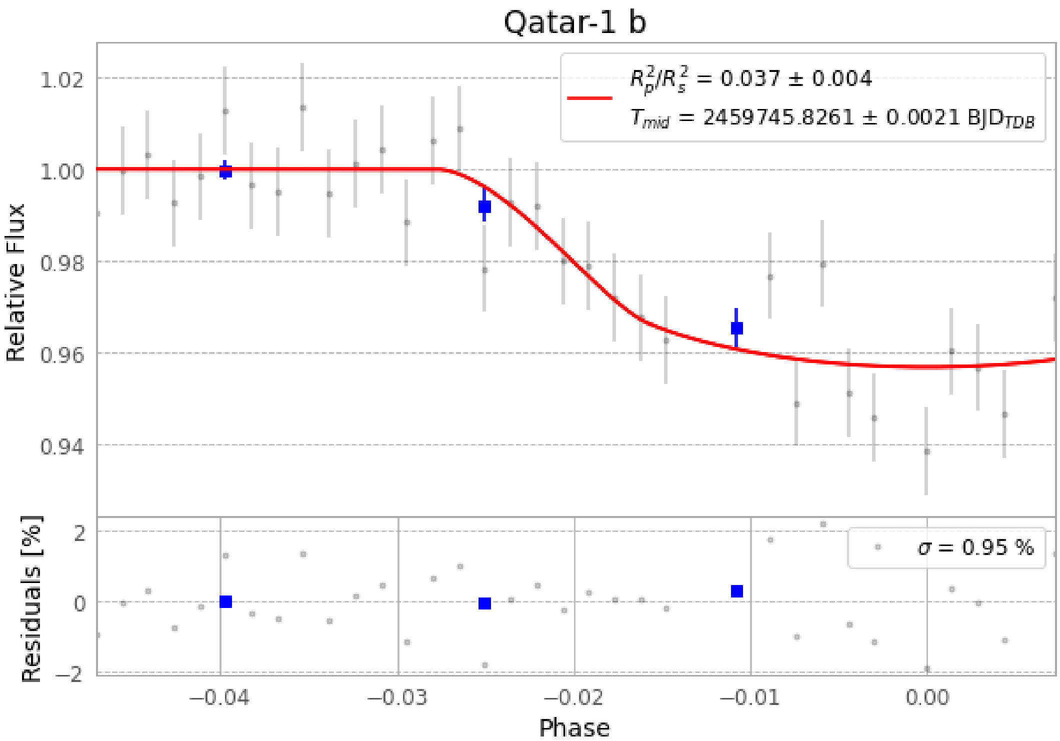

Figure 1 — FinalLightCurve_Qatar-1 b_2022-06-14.png (SHA-256 45817d40ee601e58…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [403, 366], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 0f67da6416b26753…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

34 FITS frames (22 MB), `Qatar-1220615061215.FITS` → `Qatar-1220615080315.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-220615.log (fef94cae9bf9…), FinalLightCurve_Qatar-1 b_2022-06-14.png (45817d40ee60…), FinalLightCurve_Qatar-1 b_2022-06-14.csv (80025e9e2996…)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-25 23:46Z.